Validity Analysis: FOLIO

Target context: US Undergraduate Introductory Logic Course — Automated Validity Checker

Overall risk: **HIGH**

Deployment Context

We are building an automated logic-checking tool for undergraduate formal reasoning courses at a US university. The tool receives a set of premises and a conclusion expressed in controlled natural language and determines whether the argument is logically valid, invalid, or indeterminate. Our users are American undergraduates taking introductory logic courses. We need to evaluate the LLM's deductive reasoning capabilities.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	2	Significant gaps	high
Input Content ■	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology ✓	4	Minor gaps	high
Output Content	3	Moderate gaps	high
Output Form ✓	4	Minor gaps	high
Average	**3.2**		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

FOLIO is a partially valid benchmark for a US undergraduate introductory logic course's automated validity checker. Its three-way label schema (True/False/Unknown) cleanly maps to the course rubric (valid/invalid/indeterminate) — including the course's strict 'invalid = premises entail the negation' definition, which is operationally satisfied by FOLIO's resolution-based inference engine [WEB-3, WEB-4]. The expert-annotated, FOL-engine-verified labels [Q2, Q16, Q57] provide strong ground-truth quality on the FOL portion of the course. However, two structural issues limit overall validity: (1) FOLIO contains no propositional-only subset, leaving the entire first half of the course uncovered (every sampled example uses ∀ or ∃ — DATASET-D1, D2, D4); and (2) the dominant FOLIO input register is naturalistic, Wikipedia-seeded prose with embedded real-world domain knowledge [Q35, DATASET-D7, D9, D14, D25], substantially different from the course's syntactically controlled Hurley-style templates. Reasoning depth also skews deeper than typical course problems [Q60]. Additional concerns include observable annotation errors in the HuggingFace data [DATASET-D27, D28, D29] and a paper-vs-data label-string mismatch ('Unknown' vs 'Uncertain', DATASET-D3). For a tool intended to cover the full course, FOLIO must be supplemented with a propositional-logic benchmark (LogicBench [WEB-1]) and ideally a small course-packet-aligned probe set to bridge the register gap.

ID	Evidence
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q16	“FOLIO is a high-quality and manually curated dataset, written by CS undergraduate and graduate students and researchers in academia and industry.” (p.2)
Q35	“We ask the annotators to create new stories from scratch without using templates based on real-world knowledge, which should be plausible in general.” (p.4)
Q57	“Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine.” (p.5)
Q60	“As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1.” (p.5)
D1	No plants are fungi. Mushrooms are fungi." / conclusion: "No plants are mushrooms. — Two-premise syllogism with "No A are B" / "All C are B" form — structurally close to Hurley-style templates
D2	Some cats are not pets. All cats are mammals." / conclusion: "Some mammals are not pets. — Minimal two-premise syllogism using Hurley-canonical forms ("Some A is not B", "All A are B")
D3	All proteins are organic compounds. All enzymes are organic compounds." / conclusion: "All enzymes are proteins. — Two-premise syllogism illustrating classic undistributed-middle fallacy — directly analogous to Hurley course problems
D4	Each building is tall. Everything tall has height." / conclusion: "All buildings are magnificent. — Minimal two-premise syllogism with irrelevant conclusion — indeterminate case, pedagogically transparent
D7	All people in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises. People in this tech company who wear the same flannel shirts every day are consistent... — Long naturalistic prose narrative with 7 premises and XOR embedded in premise — substantially more complex than Hurley problems
D9	All velvet-finish lipsticks in the Rouge Dior set, Lunar New Year Limited Edition are refillable...ROUGE Dior Colored Lip Balm 999 is a lipstick in the set, and it either has 'rosewood' in its official description or has a velvet finish. — Complex branded-product scenario with 5 premises and compound conclusion
D14	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency...There is an embargo on Russian foreign trade goods." / conclusion: "In Russia, an effective monetary policy is possible. — Economics-domain story; False via chain of conditionals; requires domain knowledge about Russia
D25	All orphan planets are rogue planets...PSO J318.5–22 is a rogue planet." / conclusion: "PSO J318.5–22 is an orphan planet or it does not have the Sun as its star, or both. — Astronomy domain; complex disjunctive conclusion
D27	FOL premise: "¬(∃y (flannelShirt(y) ∧ WearEveryday(x, y)) ∧ Have(mike, highEnergy) ∧ Impulsive(mike)) → (Consistent(mike) ∧ StickTo(mike, theirRegularRoutine)) ⊕ ¬Like(mike, surprise)" — Malformed FOL — free variable `x` in antecedent of conditional; annotation error
D28	Matt does not invest in the public stock market regularly. Matt likes financial risks." / conclusion: "Matt is not at risk of a gambling addiction and Mike does not both read... — Conclusion mentions "Mike" but premises are about "Matt" — name inconsistency in annotation
D29	Lily is in James' family; she watches TV series in cinemas." FOL: "Customer(lily) ∧ In(lily, jameSFamily ∧ WatchIn(lily, tV, cinema) — Missing closing parenthesis in FOL formula — syntactic annotation error
WEB-1	LogicBench uses heuristic single-rule templates but does not replicate Hurley-style bare templates (https://aclanthology.org/2024.acl-long.739/)
WEB-3	CGD-PD paper formalizes FOLIO labels as True = S ■ H, False = S ■ ¬H, Unknown = S ■ H and S ■ ¬H (https://arxiv.org/abs/2604.06196)
WEB-4	Stanford CS221 resolution-based inference engine semantics confirm False is returned only when negation is provably entailed (https://stanford-cs221.github.io/autumn2022-extra/modules/logic/first-order-resolution.pdf)

Practical Guidance

What This Benchmark Measures

FOLIO measures a model's ability to determine, from naturalistic English prose containing first-order logical structure, whether a conclusion is entailed (True), its negation is entailed (False), or neither follows (Unknown). The output ontology (Output Ontology, strong) maps directly to the course's three-way valid/invalid/indeterminate rubric. The output content (Output Content, moderate) is generally high-quality due to expert annotation and FOL-engine verification, though some annotation errors are observable in the public release. For the FOL/second-half portion of the course (covered by Input Ontology and observable in DATASET-D21, D8, D38), FOLIO provides meaningful signal at the higher end of difficulty.

Construct Depth

FOLIO probes reasoning across deep chains (mode 4 steps, 28.7% at 5+ [Q60]) and a wide AST variety [Q61], so a model scoring well on FOLIO has been tested on combinations of quantifiers, connectives, and chained inferences. However, it does not probe propositional-only reasoning at all (DATASET-D1, D2 confirm universal quantifier presence in every example) and does not probe the syntactically controlled Hurley-style register that dominates the course packet (Input Content is the weakest dimension after Input Ontology). Per-class accuracy is essential because models exhibit systematic bias toward True predictions [Q155, Q156]; overall accuracy can mask poor invalid/indeterminate performance.

What Else You Need

Three supplementation steps are necessary: (1) Add LogicBench [WEB-1] to cover propositional inference rules (MP, MT, HS, DS, etc.) for the course's first half — this directly addresses the IO gap. (2) Construct a small course-packet-aligned probe set in Hurley-style templates to address the IC register mismatch and validate model performance on the deployment's dominant register. (3) Prefer the 2026 Vampire-cleaned FOLIO split [WEB-6] over the raw HuggingFace release to mitigate the annotation errors observed in OC. Reporting per-class accuracy (especially for False and Unknown) is essential given documented model bias [Q155, Q156, Q157].

ID	Evidence
Q60	“As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1.” (p.5)
Q61	“Table 1 shows that FOLIO also has a much larger number of distinct ASTs than the previous datasets, indicating that FOLIO is much more logically diverse.” (p.5)
Q155	“Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown.” (p.15)
Q156	“The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting.” (p.15)
Q157	“They also tend to make more predictions of True than the other labels.” (p.15)
D1	No plants are fungi. Mushrooms are fungi." / conclusion: "No plants are mushrooms. — Two-premise syllogism with "No A are B" / "All C are B" form — structurally close to Hurley-style templates
D2	Some cats are not pets. All cats are mammals." / conclusion: "Some mammals are not pets. — Minimal two-premise syllogism using Hurley-canonical forms ("Some A is not B", "All A are B")
D8	A project is written either in C++ or Python. If Sam does a project written in Python, he will not use a Mac. Sam is using a Mac. If Sam uses a Mac, he will play a song. — 6-premise narrative scenario with chained conditionals and exclusive disjunction
D21	All professional athletes spend most of their time on sports...If Amy is not a Nobel physics laureate, then Amy is not an Olympic gold medal winner." / conclusion: "Amy is neither a full-time scientist nor an Olympic gold medal winner. — 6-premise chained syllogism with conditional; multi-step reasoning
D38	Harry, who lives in Potterville either yells or flies. Potter, who lives in Potterville, is a wizard and flies." / conclusion: "Harry is cool. — Fantasy-domain story; multiple conclusions from same story_id test same premises
WEB-1	LogicBench uses heuristic single-rule templates but does not replicate Hurley-style bare templates (https://aclanthology.org/2024.acl-long.739/)
WEB-6	2026 Agentified Assessment paper using Vampire theorem prover identified label errors and NL-FOL misalignments in original FOLIO; auto-formalization reached 86.70% on validation set after cleaning (https://arxiv.org/abs/2603.02788)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US Undergraduate Introductory Logic Course — Automated Validity Checker*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO's logical scope aligns with the second half of the deployment course (predicate/first-order logic with quantifiers, connectives, and combinations [Q43, Q138]), and the operator inventory comprehensively covers the FOL portion of the curriculum [DATASET-D21, D8, D38]. However, the benchmark contains no quantifier-free or propositional-only subset — every sampled example uses ∀ or ∃ [DATASET-D1, D2, D4], and the documentation treats FOL with quantifiers as the inherent design [Q15, Q63]. Since the course's first half is entirely propositional, FOLIO alone gives zero coverage of roughly half the course's logical scope. Reasoning depth also skews deeper than introductory problems: the mode is 4 steps with 28.7% requiring 5+ [Q60], whereas course-packet problems typically span 1–3 steps. Modal and temporal logic are explicitly excluded from FOLIO [Q139, Q140], which matches the deployment's classical-deductive-only constraint.

ID	Evidence
Q15	“We present a natural language reasoning dataset, FOLIO, with first-order logic reasoning problems which require the models to decide the correctness of conclusions given a world defined by the premises.” (p.1)
Q43	“There are 256 logically distinct types of syllogisms and 24 of them are valid (Lehman, 1973). We use various combinations of 24 valid syllogisms. We also add in conjunction, disjunction, and implication.” (p.4)
Q60	“As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1.” (p.5)
Q63	“We define two new tasks based on FOLIO, natural language reasoning with first-order logic and NL-FOL translation.” (p.6)
Q138	“We include the following operators: negation ¬, conjunction ∧, disjunction ∨, implication →, universal quantifier ∀, existential quantifier ∃, equal =.” (p.13)
Q139	“Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics.” (p.13)
Q140	“Consequently, they are beyond the scope of the definition of first-order logic used in our dataset.” (p.13)
D1	No plants are fungi. Mushrooms are fungi." / conclusion: "No plants are mushrooms. — Two-premise syllogism with "No A are B" / "All C are B" form — structurally close to Hurley-style templates
D2	Some cats are not pets. All cats are mammals." / conclusion: "Some mammals are not pets. — Minimal two-premise syllogism using Hurley-canonical forms ("Some A is not B", "All A are B")
D4	Each building is tall. Everything tall has height." / conclusion: "All buildings are magnificent. — Minimal two-premise syllogism with irrelevant conclusion — indeterminate case, pedagogically transparent
D8	A project is written either in C++ or Python. If Sam does a project written in Python, he will not use a Mac. Sam is using a Mac. If Sam uses a Mac, he will play a song. — 6-premise narrative scenario with chained conditionals and exclusive disjunction
D21	All professional athletes spend most of their time on sports...If Amy is not a Nobel physics laureate, then Amy is not an Olympic gold medal winner." / conclusion: "Amy is neither a full-time scientist nor an Olympic gold medal winner. — 6-premise chained syllogism with conditional; multi-step reasoning
D38	Harry, who lives in Potterville either yells or flies. Potter, who lives in Potterville, is a wizard and flies." / conclusion: "Harry is cool. — Fantasy-domain story; multiple conclusions from same story_id test same premises

Strengths

- FOL operator coverage (∀, ∃, ¬, ∧, ∨, →, =, ⊕) comprehensively matches the second-half course scope [Q138, DATASET-D21, D8]
- Explicit exclusion of modal and temporal logic [Q139, Q140] matches the course's classical-deductive-only scope

• Contains some minimal syllogistic forms structurally close to Hurley templates (e.g., 'No A are B' / 'All C are B') usable for the predicate-logic half [DATASET-D1, D2, D5]

ID	Evidence
Q138	“We include the following operators: negation ¬, conjunction ∧, disjunction ∨, implication →, universal quantifier ∀, existential quantifier ∃, equal =.” (p.13)
Q139	“Following (Russell and Norvig, 2010), we consider temporal logic and modal logic as special-purpose logics.” (p.13)
Q140	“Consequently, they are beyond the scope of the definition of first-order logic used in our dataset.” (p.13)
D1	No plants are fungi. Mushrooms are fungi." / conclusion: "No plants are mushrooms. — Two-premise syllogism with "No A are B" / "All C are B" form — structurally close to Hurley-style templates
D2	Some cats are not pets. All cats are mammals." / conclusion: "Some mammals are not pets. — Minimal two-premise syllogism using Hurley-canonical forms ("Some A is not B", "All A are B")
D5	No television stars are certified public accountants. All certified public accountants have good business sense." / conclusion: "All television stars have good business sense. — Two-premise syllogism, "No A are B" / "All B are C" form — near-Hurley template register
D8	A project is written either in C++ or Python. If Sam does a project written in Python, he will not use a Mac. Sam is using a Mac. If Sam uses a Mac, he will play a song. — 6-premise narrative scenario with chained conditionals and exclusive disjunction
D21	All professional athletes spend most of their time on sports...If Amy is not a Nobel physics laureate, then Amy is not an Olympic gold medal winner." / conclusion: "Amy is neither a full-time scientist nor an Olympic gold medal winner. — 6-premise chained syllogism with conditional; multi-step reasoning

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Deployment requires two categories: (a) propositional-logic problems (truth tables, MP, MT, HS, DS, CD, etc.) for the first half and (b) first-order predicate logic problems with quantifiers and basic syllogistic forms for the second half. The Hurley-style register is dominant across both halves.

IO-2: FOLIO omits the entire propositional-only category. Every sampled example contains at least one quantifier [DATASET-D1, D2, D4], and the documentation treats FOL with quantifiers as foundational [Q15, Q63]. LogicBench [WEB-1] is identified as a complementary benchmark covering nine propositional inference rules.

IO-3: FOLIO includes very deep reasoning chains (mode 4 steps, 28.7% requiring 5+ [Q60]; DATASET-D7, D10, D21 show 6–7 premise chains) that exceed typical introductory difficulty and may introduce construct-irrelevant variance for the deployment's 1–3 step target.

IO-4: Two documented gaps harm content validity: (a) total absence of propositional-only subset undertests the first half of the course; (b) reasoning depth distribution skews well above the typical course-packet 1–3 step range.

ID	Evidence
Q15	“We present a natural language reasoning dataset, FOLIO, with first-order logic reasoning problems which require the models to decide the correctness of conclusions given a world defined by the premises.” (p.1)
Q43	“There are 256 logically distinct types of syllogisms and 24 of them are valid (Lehman, 1973). We use various combinations of 24 valid syllogisms. We also add in conjunction, disjunction, and implication.” (p.4)
Q60	“As shown in Figure 1, the mode of reasoning depths is four in FOLIO. 28.7% of the examples need five or more depths of reasoning to infer the conclusions, while the previous datasets needed at most five reasoning depths as shown in Table 1.” (p.5)

Q63	“We define two new tasks based on FOLIO, natural language reasoning with first-order logic and NL-FOL translation.” (p.6)
D1	No plants are fungi. Mushrooms are fungi." / conclusion: "No plants are mushrooms. — Two-premise syllogism with "No A are B" / "All C are B" form — structurally close to Hurley-style templates
D2	Some cats are not pets. All cats are mammals." / conclusion: "Some mammals are not pets. — Minimal two-premise syllogism using Hurley-canonical forms ("Some A is not B", "All A are B")
D4	Each building is tall. Everything tall has height." / conclusion: "All buildings are magnificent. — Minimal two-premise syllogism with irrelevant conclusion — indeterminate case, pedagogically transparent
D7	All people in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises. People in this tech company who wear the same flannel shirts every day are consistent... — Long naturalistic prose narrative with 7 premises and XOR embedded in premise — substantially more complex than Hurley problems
D10	All people who attend Renaissance fairs regularly enjoy dressing up in old-fashioned and historical period clothing...If Clyde is not focused on futuristic and vocational subjects, then he is neither focused on futuristic and vocational subjects nor enjoys dressing up... — 6-premise Wikipedia-seeded narrative; convoluted tautological final premise
D21	All professional athletes spend most of their time on sports...If Amy is not a Nobel physics laureate, then Amy is not an Olympic gold medal winner." / conclusion: "Amy is neither a full-time scientist nor an Olympic gold medal winner. — 6-premise chained syllogism with conditional; multi-step reasoning
WEB-1	LogicBench uses heuristic single-rule templates but does not replicate Hurley-style bare templates (https://aclanthology.org/2024.acl-long.739/)
WEB-2	FOL2NS characterizes quantifiers as the key feature distinguishing FOL from propositional logics; FOLIO has no propositional-only subset (https://arxiv.org/abs/2605.18155)

Information Gaps

- Exact proportion of FOLIO examples that fall in the 1–3 step depth range usable as proxy for course-packet difficulty

Requires Expert Verification

- Course instructor confirmation of typical reasoning depth distribution in actual course packets

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Remediation

Gap 1: Complete absence of propositional-only (quantifier-free) examples leaves the course's first half uncovered.

Recommendation: Supplement FOLIO with LogicBench [WEB-1], which explicitly covers nine propositional inference rules (MP, MT, HS, DS, CD, DD, BD, CT, MI) directly relevant to the propositional half of the course.

Gap 2: Reasoning depth distribution (mode 4 steps, 28.7% at 5+) substantially exceeds typical introductory course-packet problems (1–3 steps).

Recommendation: Stratify FOLIO evaluation by reasoning depth and report 1–3 step accuracy separately as the course-aligned metric; treat 5+ step performance as informative but not deployment-critical.

ID	Evidence
WEB-1	LogicBench uses heuristic single-rule templates but does not replicate Hurley-style bare templates (https://aclanthology.org/2024.acl-long.739/)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US Undergraduate Introductory Logic Course — Automated Validity Checker*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The deployment's dominant input register is heavily templated Hurley-style prose ('All A are B', 'If P then Q'), explicitly confirmed in elicitation A3. FOLIO's WikiLogic pipeline deliberately avoided template-style generation [Q35] and produces naturalistic, Wikipedia-seeded prose; HybLogic uses syllogistic templates instantiated with naturalistic NL fillers [Q42, Q44]. Sampled data confirms predominance of complex naturalistic narratives — tech companies [DATASET-D7], branded products [DATASET-D9], economics with geopolitical content [DATASET-D14], astronomy with technical nomenclature [DATASET-D25]. Only a minority of examples [DATASET-D1, D2, D3, D5] approach Hurley-template form. FOLIO also added epistemic qualifiers to generalizations [Q129] and used naturalistic disjunction phrasings [Q130, Q131, Q132, Q133] that depart from bare templates. WikiLogic content may overlap with LLM training data [Q104], further confounding deployment performance prediction. The benchmark does explicitly avoid identity-linked biases and stereotypes [Q47, Q127], which is appropriate for a US academic context.

ID	Evidence
Q35	“We ask the annotators to create new stories from scratch without using templates based on real-world knowledge, which should be plausible in general.” (p.4)
Q42	“In this approach, we first generate logically valid stories, which are templates containing abstract categories and entities, by combining multiple syllogisms into a single story template: the conclusion of one syllogism is used as a premise for the next syllogism.” (p.4)
Q44	“We then ask human annotators to assign nouns, phrases, or clauses to the abstract entities or categories that reflect real-life scenarios to each template and write logically-valid stories in natural language.” (p.4)
Q47	“Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion.” (p.4)
Q104	“Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data.” (p.8)
Q127	“We took out stories that had strongly opinionated language and contained gender, racial, and classist biases.” (p.13)
Q129	“For stories that pertain to generalization, such as "All As are Bs," we have added specifiers like "all Dan knows" to give a degree of reasonable factuality.” (p.13)
Q130	“We always use "either-or" to express exclusive disjunction.” (p.13)
Q131	“We use either "A or B" or "A or B, or both" to express inclusive disjunction.” (p.13)
Q132	“It is more natural to say "Some A is B" rather than "there exists an A such that A is B."” (p.13)
Q133	“"All A are B" can be more natural than "If A then B".” (p.13)
D1	No plants are fungi. Mushrooms are fungi." / conclusion: "No plants are mushrooms. — Two-premise syllogism with "No A are B" / "All C are B" form — structurally close to Hurley-style templates
D2	Some cats are not pets. All cats are mammals." / conclusion: "Some mammals are not pets. — Minimal two-premise syllogism using Hurley-canonical forms ("Some A is not B", "All A are B")
D3	All proteins are organic compounds. All enzymes are organic compounds." / conclusion: "All enzymes are proteins. — Two-premise syllogism illustrating classic undistributed-middle fallacy — directly analogous to Hurley course problems

D5	No television stars are certified public accountants. All certified public accountants have good business sense." / conclusion: "All television stars have good business sense. — Two-premise syllogism, "No A are B" / "All B are C" form — near-Hurley template register
D7	All people in this tech company who are consistent and enjoy sticking to their regular routines do not like surprises. People in this tech company who wear the same flannel shirts every day are consistent... — Long naturalistic prose narrative with 7 premises and XOR embedded in premise — substantially more complex than Hurley problems
D9	All velvet-finish lipsticks in the Rouge Dior set, Lunar New Year Limited Edition are refillable...ROUGE Dior Colored Lip Balm 999 is a lipstick in the set, and it either has 'rosewood' in its official description or has a velvet finish. — Complex branded-product scenario with 5 premises and compound conclusion
D14	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency...There is an embargo on Russian foreign trade goods." / conclusion: "In Russia, an effective monetary policy is possible. — Economics-domain story; False via chain of conditionals; requires domain knowledge about Russia
D25	All orphan planets are rogue planets...PSO J318.5–22 is a rogue planet." / conclusion: "PSO J318.5–22 is an orphan planet or it does not have the Sun as its star, or both. — Astronomy domain; complex disjunctive conclusion

Strengths

- Subset of HybLogic examples uses minimal syllogistic forms structurally close to Hurley templates [DATASET-D1, D2, D5]
- Explicit exclusion of identity stereotypes and opinionated language [Q47, Q127] matches academic deployment norms
- Topic diversity and AST variety [Q38, Q61] reduce risk of narrow-domain overfit

ID	Evidence
Q38	“This results in a wide range of topics, abundant AST variations, and a wide vocabulary for FOLIO.” (p.4)
Q47	“Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion.” (p.4)
Q61	“Table 1 shows that FOLIO also has a much larger number of distinct ASTs than the previous datasets, indicating that FOLIO is much more logically diverse.” (p.5)
Q127	“We took out stories that had strongly opinionated language and contained gender, racial, and classist biases.” (p.13)
D1	No plants are fungi. Mushrooms are fungi." / conclusion: "No plants are mushrooms. — Two-premise syllogism with "No A are B" / "All C are B" form — structurally close to Hurley-style templates
D2	Some cats are not pets. All cats are mammals." / conclusion: "Some mammals are not pets. — Minimal two-premise syllogism using Hurley-canonical forms ("Some A is not B", "All A are B")
D5	No television stars are certified public accountants. All certified public accountants have good business sense." / conclusion: "All television stars have good business sense. — Two-premise syllogism, "No A are B" / "All B are C" form — near-Hurley template register

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Deployment requires syntactically constrained, domain-neutral pedagogical templates. FOLIO inputs frequently embed real-world domain knowledge (Russia/embargo D14, Rouge Dior products D9, PSO J318.5–22 D25) that introduces construct-irrelevant variance not present in Hurley-style problems.

IC-2: Cultural content is broadly US/Western academic (Wikipedia-seeded). Aligns with US undergraduate deployment population. No identity-linked biases were retained [Q47, Q127].

IC-3: Western-specific knowledge appears (Yale references in D11, US tech companies in D23, US film industry in D12) but the deployment is itself US-based, so this is not a transfer problem. The greater concern is construct-irrelevant domain knowledge load.

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotator recruitment was performed for this assessment; concern is based on register/style mismatch, not cultural sensitivity.

IC-5: Register mismatch (naturalistic prose vs. Hurley-style bare templates) is the primary content-validity concern; LLM training-data contamination for WikiLogic [Q104] adds a secondary concern.

ID	Evidence
Q35	“We ask the annotators to create new stories from scratch without using templates based on real-world knowledge, which should be plausible in general.” (p.4)
Q47	“Our dataset prioritizes realism and factual accuracy, steering clear of biases and stereotypes linked to identity markers like race, ethnicity, gender, sexuality, nationality, class, and religion.” (p.4)
Q104	“Moreover, since the examples in WikiLogic are created from scratch by humans, it is possible that LLMs have seen similar texts with similar logical patterns in the training data.” (p.8)
Q127	“We took out stories that had strongly opinionated language and contained gender, racial, and classist biases.” (p.13)
Q129	“For stories that pertain to generalization, such as "All As are Bs," we have added specifiers like "all Dan knows" to give a degree of reasonable factuality.” (p.13)
D9	All velvet-finish lipsticks in the Rouge Dior set, Lunar New Year Limited Edition are refillable...ROUGE Dior Colored Lip Balm 999 is a lipstick in the set, and it either has 'rosewood' in its official description or has a velvet finish. — Complex branded-product scenario with 5 premises and compound conclusion
D11	No trick-shot artist in Yale's varsity team struggles with half court shots...Jack is on Yale's varsity team, and he is either a trick-shot artist or he successfully shoots a high percentage of 3-pointers. — Sports-domain narrative, 5 premises, label=False — tests invalid-conclusion case
D12	Michael O'Donnell is a British physician, journalist, author, and broadcaster...Michael O'Donnell was born in Yorkshire as the son of a general practitioner." / conclusion: "There are no journalists that were born in Yorkshire. — WikiLogic biographical narrative; False label from clear counterexample in premises
D14	For a country, if effective monetary policy is possible, it must have successful inflation control and a strong national currency...There is an embargo on Russian foreign trade goods." / conclusion: "In Russia, an effective monetary policy is possible. — Economics-domain story; False via chain of conditionals; requires domain knowledge about Russia
D23	Everyone working at Meta has a high income...James has a car or works at Meta." / conclusion: "James is a student. — Modern tech-company scenario; False conclusion derivable via short chain
D25	All orphan planets are rogue planets...PSO J318.5–22 is a rogue planet." / conclusion: "PSO J318.5–22 is an orphan planet or it does not have the Sun as its star, or both. — Astronomy domain; complex disjunctive conclusion
WEB-1	LogicBench uses heuristic single-rule templates but does not replicate Hurley-style bare templates (https://aclanthology.org/2024.acl-long.739/)

Information Gaps

- Proportion of HybLogic examples that match Hurley templates closely enough to use as a focused subset
- Quantitative measure of LLM training-data contamination risk for WikiLogic examples

Requires Expert Verification

- Course-instructor sampling of representative course-packet problems to compare against FOLIO subsets for register match

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Remediation

Gap 1: FOLIO's dominant naturalistic prose register (with Wikipedia-seeded content and real-world domain knowledge) does not match the course's Hurley-style syntactically controlled templates.

Recommendation: Construct a small course-packet-aligned probe set (50–100 problems) in bare Hurley-template form ('All A are B', 'If P then Q', etc.) for register-matched validation. Optionally filter FOLIO to the HybLogic subset that approaches template form (e.g., DATASET-D1, D2, D5) for a closer FOLIO-internal proxy.

ID	Evidence
D1	No plants are fungi. Mushrooms are fungi." / conclusion: "No plants are mushrooms. — Two-premise syllogism with "No A are B" / "All C are B" form — structurally close to Hurley-style templates
D2	Some cats are not pets. All cats are mammals." / conclusion: "Some mammals are not pets. — Minimal two-premise syllogism using Hurley-canonical forms ("Some A is not B", "All A are B")
D5	No television stars are certified public accountants. All certified public accountants have good business sense." / conclusion: "All television stars have good business sense. — Two-premise syllogism, "No A are B" / "All B are C" form — near-Hurley template register

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US Undergraduate Introductory Logic Course — Automated Validity Checker*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Modality, script, and language match the deployment cleanly: both are English text in standard Latin script [Q27, Q64]. FOLIO's premises-and-conclusions structural format is directly compatible with the deployment's input style. Grammar was checked with Grammarly and reviewed by English Literature annotators [Q49, Q50], and ambiguity was minimized [Q51]. The 70/15/15 train/val/test split by story [Q78, Q79] avoids cross-split leakage. The only form-level wrinkles are: (a) FOLIO uses naturalistic disjunction conventions [Q130, Q131] and adds epistemic qualifiers [Q129] rather than bare connective phrasings — but these are register-level (input content) rather than modality-level concerns; (b) the HuggingFace dataset's mean Dale-Chall readability score could not be retrieved [WEB-7]. No infrastructure or signal-distribution mismatch.

ID	Evidence
Q27	“Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.3)
Q49	“Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly.” (p.4)
Q50	“Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness.” (p.4)
Q51	“We also eliminate natural language ambiguity when it is possible.” (p.4)
Q64	“Each natural language (NL) story S in FOLIO consists of n premises: P = {p1, p2, ..., pn} and m conclusions: H = {h1, h2, ..., hm}.” (p.6)
Q78	“We split FOLIO by 70%/15%/15% split for the train/validation/test sets with 1,001/203/226 examples respectively.” (p.6)
Q79	“We split by story so that models are evaluated on unseen stories.” (p.6)
Q129	“For stories that pertain to generalization, such as "All As are Bs," we have added specifiers like "all Dan knows" to give a degree of reasonable factuality.” (p.13)
Q130	“We always use "either-or" to express exclusive disjunction.” (p.13)
Q131	“We use either "A or B" or "A or B, or both" to express inclusive disjunction.” (p.13)
WEB-7	FOLIO GitHub repository — Figure 3 readability distribution not retrievable from text passages; mean Dale-Chall not reported (https://github.com/Yale-LILY/FOLIO)

Strengths

- Text-only English in standard Latin script matches deployment exactly [Q27, Q64]
- Grammar-checked, ambiguity-minimized prose [Q49, Q50, Q51]
- Story-level train/val/test split prevents leakage [Q78, Q79]

ID	Evidence
Q27	“Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.3)
Q49	“Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly.” (p.4)

Q50	“Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness.” (p.4)
Q51	“We also eliminate natural language ambiguity when it is possible.” (p.4)
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Q78	“We split FOLIO by 70%/15%/15% split for the train/validation/test sets with 1,001/203/226 examples respectively.” (p.6)
Q79	“We split by story so that models are evaluated on unseen stories.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: No signal-distribution mismatch — both benchmark and deployment use English text. Dale-Chall readability distribution is reported in FOLIO Figure 3 [Q154] but no mean is accessible; Hurley templates likely score lower (simpler) than FOLIO's naturalistic prose [WEB-7].

IF-2: Deployment infrastructure is standard US-university NLP tooling; full compatibility with FOLIO's text format. NLI tooling availability is strong for English.

IF-3: Sentence-level conventions (e.g., 'Some A is B' over '∃x...' [Q132], 'All A are B' over 'If A then B' [Q133]) approach Hurley forms; 'either-or' for XOR [Q130] is also a textbook convention. Where FOLIO diverges is in qualifier addition [Q129] and richer naturalistic prose.

IF-4: No modality or infrastructure mismatch identified. Residual readability concern flagged but low-priority.

ID	Evidence
Q27	“Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.3)
Q49	“Apart from grammar, we make sure the sentences in our dataset are highly natural. All the sentences are first checked with a grammar checking tool, Grammarly.” (p.4)
Q50	“Our annotators who have graduated from or are senior students studying English Literature conducted a thorough round of review for grammatical correctness and language naturalness.” (p.4)
Q64	“Each natural language (NL) story S in FOLIO consists of n premises: P = {p1, p2, ..., pn} and m conclusions: H = {h1, h2, ..., hm}.” (p.6)
Q129	“For stories that pertain to generalization, such as "All As are Bs," we have added specifiers like "all Dan knows" to give a degree of reasonable factuality.” (p.13)
Q130	“We always use "either-or" to express exclusive disjunction.” (p.13)
Q132	“It is more natural to say "Some A is B" rather than "there exists an A such that A is B."” (p.13)
Q133	“"All A are B" can be more natural than "If A then B".” (p.13)
Q154	“We show the distribution of readability in Figure 3.” (p.14)
WEB-7	FOLIO GitHub repository — Figure 3 readability distribution not retrievable from text passages; mean Dale-Chall not reported (https://github.com/Yale-LILY/FOLIO)

Information Gaps

- Mean Dale-Chall readability score for FOLIO vs. typical course packet

Output Ontology (OO)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US Undergraduate Introductory Logic Course — Automated Validity Checker*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

FOLIO uses a three-way True/False/Unknown label schema [Q36, Q66] that maps onto the deployment's valid/invalid/indeterminate rubric. The label-mapping concern flagged in elicitation (whether 'False' = entailment of negation, as the course requires for 'invalid') is resolved as aligned: the proving pipeline returns a definite truth value via theorem-prover execution [Q153], and the resolution-based Stanford CS221 engine returns False only when the negation is provably entailed [WEB-3, WEB-4]. The Unknown label is formally defined as the case where the inference engine returns neither True nor False [Q70], which matches the course's indeterminate definition (S ■ H and S ■ ¬H). Undecidable formulas are excluded by construction [Q70], matching the course rubric's exclusion. Empirical inspection confirms False labels are assigned only with provable counterexamples [DATASET-D11, D12, D13]. Two residual concerns: (a) the HuggingFace dataset uses the label string 'Uncertain' rather than the paper's 'Unknown' [DATASET-D3, D15, D37], a practical-implementation hazard; (b) label imbalance (Unknown = 27% of test set [Q94]) combined with model bias toward True predictions [Q155, Q156, Q157] means accuracy may not equally reflect performance on the invalid and indeterminate categories the rubric distinguishes.

ID	Evidence
Q36	“Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example).” (p.4)
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q70	“In our dataset, the premises and conclusions are set up in such a way to ensure that the inference engine always returns an answer given enough resources such as time and memory.” (p.6)
Q94	“The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively.” (p.7)
Q153	“Proving a story requires three steps. First, the FOL statements of the premises and conclusions of a story annotated by humans are converted to Python code. Then, the code snippets are used as input to the theorem prover. Finally, the theorem prover outputs whether the conclusions are True / False / Unknown, based on the premises.” (p.14)
Q155	“Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown.” (p.15)
Q156	“The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting.” (p.15)
Q157	“They also tend to make more predictions of True than the other labels.” (p.15)
D3	All proteins are organic compounds. All enzymes are organic compounds." / conclusion: "All enzymes are proteins. — Two-premise syllogism illustrating classic undistributed-middle fallacy — directly analogous to Hurley course problems
D11	No trick-shot artist in Yale's varsity team struggles with half court shots...Jack is on Yale's varsity team, and he is either a trick-shot artist or he successfully shoots a high percentage of 3-pointers. — Sports-domain narrative, 5 premises, label=False — tests invalid-conclusion case
D12	Michael O'Donnell is a British physician, journalist, author, and broadcaster...Michael O'Donnell was born in Yorkshire as the son of a general practitioner." / conclusion: "There are no journalists that were born in Yorkshire. — WikiLogic biographical narrative; False label from clear counterexample in premises
D13	The only types of mammals that lay eggs are either platypuses or echidnas...Grebes are not platypuses and also not echidnas." / conclusion: "Hyraxes lay eggs. — Biology-domain story; False conclusion via FOL entailment of negation

D15	Everyone at the business conference is either an investor or an entrepreneur...Ho is at the business conference and prefers state ownership of the means of production." / conclusion: "Ho is not an ardent communist. — Unknown because ardent communist status is underdetermined given premises
D37	All young adults at the event like independence...If Susan is a Yale student, then she does not like independence." / conclusion: "Susan is a college student. — 7-premise story with competing conditional chains; classic indeterminate case
WEB-3	CGD-PD paper formalizes FOLIO labels as True = S ■ H, False = S ■ ¬H, Unknown = S ■ H and S ■ ¬H (https://arxiv.org/abs/2604.06196)
WEB-4	Stanford CS221 resolution-based inference engine semantics confirm False is returned only when negation is provably entailed (https://stanford-cs221.github.io/autumn2022-extra/modules/logic/first-order-resolution.pdf)

Strengths

- Three-way schema [Q36, Q66] structurally matches deployment rubric
- False label formally equivalent to S ■ ¬H via resolution prover [WEB-3, WEB-4], matching course's strict 'invalid' definition
- Unknown excludes undecidable formulas by construction [Q70], matching course's exclusion
- Empirical False examples reflect provable negation entailment [DATASET-D11, D12, D13]

ID	Evidence
Q36	“Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example).” (p.4)
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q70	“In our dataset, the premises and conclusions are set up in such a way to ensure that the inference engine always returns an answer given enough resources such as time and memory.” (p.6)
D11	No trick-shot artist in Yale's varsity team struggles with half court shots...Jack is on Yale's varsity team, and he is either a trick-shot artist or he successfully shoots a high percentage of 3-pointers. — Sports-domain narrative, 5 premises, label=False — tests invalid-conclusion case
D12	Michael O'Donnell is a British physician, journalist, author, and broadcaster...Michael O'Donnell was born in Yorkshire as the son of a general practitioner." / conclusion: "There are no journalists that were born in Yorkshire. — WikiLogic biographical narrative; False label from clear counterexample in premises
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WEB-4	Stanford CS221 resolution-based inference engine semantics confirm False is returned only when negation is provably entailed (https://stanford-cs221.github.io/autumn2022-extra/modules/logic/first-order-resolution.pdf)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: All three FOLIO categories are deployment-relevant: True↔valid, False↔invalid (both = entailment of negation), Unknown↔indeterminate [Q36, Q66, Q70, WEB-3, WEB-4].

OO-2: No regionally-specific category missing — the deployment's three-way scheme is fully represented.

OO-3: No non-regional values encoded; FOL semantics are universal.

OO-4: No taxonomy redesign required. However, evaluation code must use the actual label string 'Uncertain' (HuggingFace) rather than 'Unknown' (paper) [DATASET-D3, D15, D37].

OO-5: Residual issues: label-string inconsistency between paper and HuggingFace data; label imbalance (Unknown 27%) combined with documented model bias toward True [Q155, Q156] reduces accuracy's informativeness for the indeterminate category.

ID	Evidence
Q36	“Each of the stories is composed of several premises and conclusions with truth values of True, False, or Unknown (see Table 2 for an example).” (p.4)
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q70	“In our dataset, the premises and conclusions are set up in such a way to ensure that the inference engine always returns an answer given enough resources such as time and memory.” (p.6)
Q94	“The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively.” (p.7)
Q138	“We include the following operators: negation ¬, conjunction ∧, disjunction ∨, implication →, universal quantifier ∀, existential quantifier ∃, equal =.” (p.13)
Q155	“Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown.” (p.15)
Q156	“The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting.” (p.15)
D3	All proteins are organic compounds. All enzymes are organic compounds." / conclusion: "All enzymes are proteins. — Two-premise syllogism illustrating classic undistributed-middle fallacy — directly analogous to Hurley course problems
D15	Everyone at the business conference is either an investor or an entrepreneur...Ho is at the business conference and prefers state ownership of the means of production." / conclusion: "Ho is not an ardent communist. — Unknown because ardent communist status is underdetermined given premises
D37	All young adults at the event like independence...If Susan is a Yale student, then she does not like independence." / conclusion: "Susan is a college student. — 7-premise story with competing conditional chains; classic indeterminate case
WEB-3	CGD-PD paper formalizes FOLIO labels as True = S ■ H, False = S ■ ¬H, Unknown = S ■ H and S ■ ¬H (https://arxiv.org/abs/2604.06196)
WEB-4	Stanford CS221 resolution-based inference engine semantics confirm False is returned only when negation is provably entailed (https://stanford-cs221.github.io/autumn2022-extra/modules/logic/first-order-resolution.pdf)

Information Gaps

- Whether FOLIO's 27% Unknown share matches the course packet's indeterminate problem frequency

Requires Expert Verification

- Course-instructor elicitation of typical course-packet indeterminate problem rate

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Remediation

Gap 1: Label string in HuggingFace data ('Uncertain') differs from paper documentation ('Unknown'); label imbalance plus documented model bias toward True predictions reduces informativeness of overall accuracy.

Recommendation: Use the actual label strings as released ('Uncertain') in evaluation pipelines, and report per-class accuracy (True/False/Uncertain separately) in addition to overall accuracy. Track invalid- and indeterminate-class accuracy specifically as the deployment-critical metrics.

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US Undergraduate Introductory Logic Course — Automated Validity Checker*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth label generation is rigorous: expert annotators (CS undergraduates/graduates and senior researchers with formal FOL training and native or near-native English [Q16, Q27, Q28, Q31, Q122, Q123]) executed a six-stage 980-person-hour annotation pipeline [Q32] with separate NL and FOL expert quality checks [Q29, Q124, Q125]. All FOL annotations are verified by an inference engine [Q2, Q57], and expert human accuracy reaches 95.98% [Q111] — strong convergent evidence for label correctness on clear cases. Premise ordering does not yield spurious labels [Q158, Q159, Q160]. However, empirical inspection of the HuggingFace data reveals multiple concrete annotation errors: free variables in FOL conditionals [DATASET-D27], name inconsistencies between premises and conclusions [DATASET-D24, D28], missing closing parentheses [DATASET-D29, D30, D35], and a potential NL-FOL alignment gap on inclusive disjunction [DATASET-D36]. The 2026 Vampire-prover cleaning study confirms these are systemic [WEB-6]. The annotator population (expert FOL-trained academics) is appropriate for producing reliable ground-truth labels but is not representative of novice-student production norms — this is acceptable for ground-truth quality and not a concern for the deployment's use of FOLIO as a label oracle.

ID	Evidence
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q16	“FOLIO is a high-quality and manually curated dataset, written by CS undergraduate and graduate students and researchers in academia and industry.” (p.2)
Q27	“Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.3)
Q28	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-” (p.3)
Q29	“At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved.” (p.4)
Q31	“All stories and FOL annotations in FOLIO are written and reviewed by expert annotators, including CS undergraduate and graduate students, and senior researchers, who met the aforementioned criteria.” (p.4)
Q32	“We develop our dataset in six stages: WikiLogic collection, HybLogic collection, NL quality control, FOL quality control, NL-FOL alignment and FOL verification, spending 980 man-hours in total.” (p.4)
Q57	“Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine.” (p.5)
Q111	“Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%.” (p.9)
Q122	“Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.12)
Q123	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing.” (p.12)
Q124	“At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved.” (p.12)
Q125	“For the FOL quality check, only annotators who are experts in first-order logic are involved.” (p.12)
Q158	“To test if the premise ordering in FOLIO has spurious correlations with the conclusion label which a model can exploit, we shuffle the input premises to evaluate models.” (p.15)

Q159	“We find that accuracy increases or decreases by roughly 1% in most settings compared to our unshuffled premises.” (p.15)
Q160	“This indicates that the ordering of premises in FOLIO examples does not yield significant information about the label, and thus models will not be able to use the premise ordering as a strong heuristic or statistical feature for its predictions.” (p.15)
D24	"No athletes never exercise...Either John is a professional basketball player and he never exercises, or he is not a professional basketball player and he sometimes exercises." (Note: conclusion uses "Jim" while premises use "John/Jim" inconsistently) — Name inconsistency between premises (John) and conclusion (Jim) — annotation quality issue
D27	FOL premise: "¬(∃y (flannelShirt(y) ∧ WearEveryday(x, y)) ∧ Have(mike, highEnergy) ∧ Impulsive(mike)) → (Consistent(mike) ∧ StickTo(mike, theirRegularRoutine)) ⊕ ¬Like(mike, surprise)" — Malformed FOL — free variable `x` in antecedent of conditional; annotation error
D28	Matt does not invest in the public stock market regularly. Matt likes financial risks." / conclusion: "Matt is not at risk of a gambling addiction and Mike does not both read... — Conclusion mentions "Mike" but premises are about "Matt" — name inconsistency in annotation
D29	Lily is in James' family; she watches TV series in cinemas." FOL: "Customer(lily) ∧ In(lily, jameSFamily ∧ WatchIn(lily, tV, cinema) — Missing closing parenthesis in FOL formula — syntactic annotation error
D30	TikTok is a social media application, and it is not ideal for preteens." / conclusion-FOL: "Contain(tikTok, chatFeature) ⊕ ComputerProgram(tikTok)) — Trailing unmatched parenthesis in conclusion-FOL
D35	(Knows(dan, dune) ∧ ScienceFiction(dune)) ∨ ProvedToBe(dune, false)) — Unmatched parenthesis in FOL formula — additional annotation error instance
D36	Someone is either a seasoned software engineer interviewer at Google, has human rights, or both." / premises-FOL: "∀x ((Seasoned(x) ∧ SoftwareEngineerInterviewer(x) ∧ At(x, google)) ∨ Have(x, humanRights)) — NL says "or both" (inclusive disjunction) but FOL uses ∨ — potential NL-FOL alignment gap
WEB-6	2026 Agentified Assessment paper using Vampire theorem prover identified label errors and NL-FOL misalignments in original FOLIO; auto-formalization reached 86.70% on validation set after cleaning (https://arxiv.org/abs/2603.02788)

Strengths

- Expert annotator population with formal FOL training [Q16, Q27, Q28, Q123]
- Six-stage 980-person-hour annotation pipeline with separate NL and FOL quality checks [Q29, Q32, Q124, Q125]
- FOL-engine verification of all label annotations [Q2, Q57]
- Expert human accuracy 95.98% [Q111] — strong convergent evidence
- No premise-ordering spurious correlations [Q158, Q159, Q160]

ID	Evidence
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q16	“FOLIO is a high-quality and manually curated dataset, written by CS undergraduate and graduate students and researchers in academia and industry.” (p.2)
Q27	“Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.3)
Q28	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-” (p.3)
Q29	“At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved. For the FOL quality check, only annotators who are experts in first-order logic are involved.” (p.4)
Q32	“We develop our dataset in six stages: WikiLogic collection, HybLogic collection, NL quality control, FOL quality control, NL-FOL alignment and FOL verification, spending 980 man-hours in total.” (p.4)
Q57	“Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine.” (p.5)
Q111	“Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%.” (p.9)
Q123	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing.” (p.12)
Q124	“At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved.” (p.12)
Q125	“For the FOL quality check, only annotators who are experts in first-order logic are involved.” (p.12)

Q158	“To test if the premise ordering in FOLIO has spurious correlations with the conclusion label which a model can exploit, we shuffle the input premises to evaluate models.” (p.15)
Q159	“We find that accuracy increases or decreases by roughly 1% in most settings compared to our unshuffled premises.” (p.15)
Q160	“This indicates that the ordering of premises in FOLIO examples does not yield significant information about the label, and thus models will not be able to use the premise ordering as a strong heuristic or statistical feature for its predictions.” (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels reflect formal FOL semantics, which is the deployment's normative standard for valid/invalid/indeterminate. Regional stakeholder perspective is replaced by formal logical correctness — appropriate for this deployment.

OC-2: Disagreement between annotators and regional population is not a meaningful axis here, since the rubric is formal-logical. Novice students would disagree with expert labels but only because the experts are correct.

OC-3: Annotator demographics documented: native/near-native English speakers, college and graduate CS students with formal FOL training, NLP and FOL domain experts assigned to quality-check stages [Q27, Q28, Q31, Q122, Q123, Q124, Q125].

OC-4: Re-annotation by regional pool not necessary — formal-logical labels are the target. However, the 2026 Vampire-prover cleaned split [WEB-6] should be preferred if available, given the annotation errors observed in the raw HuggingFace data [DATASET-D27, D28, D29, D30, D35].

OC-5: Aggregation method (FOL-engine verification) does not erase minority perspectives; it enforces formal correctness.

OC-6: Concrete annotation errors observable in sampled data (free variables, name mismatches, unmatched parentheses) confirm the 2026 cleaning study's findings — undermines label reliability for some portion of the dataset and warrants using the cleaned split.

ID	Evidence
Q2	“The logical correctness of the premises and conclusions is ensured by their FOL annotations, which are automatically verified by an FOL inference engine.” (p.1)
Q27	“Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.3)
Q28	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or seman-” (p.3)
Q31	“All stories and FOL annotations in FOLIO are written and reviewed by expert annotators, including CS undergraduate and graduate students, and senior researchers, who met the aforementioned criteria.” (p.4)
Q57	“Recognizing that the FOL formula annotations can be error-prone, we verify the syntactic validity and label consistency of FOL formula annotations with an FOL inference engine.” (p.5)
Q111	“Expert annotations achieve an accuracy of 95.98% while non-expert annotations achieves 61.82%, with a gap of 34.16%.” (p.9)
Q122	“Given the complexities of our annotations, we selected annotators based on a few important criteria 1). Our annotators are either college or graduate students who are native English speakers or possess near-native proficiency in English.” (p.12)

Q123	“They possess formal education in first-order logic, having either completed relevant coursework or undertaken self-directed studies in first-order logic or semantic parsing.” (p.12)
Q124	“At the NL quality check stage, only annotators who are experts in natural language processing or computational linguistics are involved.” (p.12)
Q125	“For the FOL quality check, only annotators who are experts in first-order logic are involved.” (p.12)
D27	FOL premise: "¬(∃y (flannelShirt(y) ∧ WearEveryday(x, y)) ∧ Have(mike, highEnergy) ∧ Impulsive(mike)) → (Consistent(mike) ∧ StickTo(mike, theirRegularRoutine)) ⊕ ¬Like(mike, surprise)" — Malformed FOL — free variable `x` in antecedent of conditional; annotation error
D28	Matt does not invest in the public stock market regularly. Matt likes financial risks." / conclusion: "Matt is not at risk of a gambling addiction and Mike does not both read... — Conclusion mentions "Mike" but premises are about "Matt" — name inconsistency in annotation
D29	Lily is in James' family; she watches TV series in cinemas." FOL: "Customer(lily) ∧ In(lily, jameSFamily ∧ WatchIn(lily, tV, cinema) — Missing closing parenthesis in FOL formula — syntactic annotation error
D30	TikTok is a social media application, and it is not ideal for preteens." / conclusion-FOL: "Contain(tikTok, chatFeature) ⊕ ComputerProgram(tikTok)) — Trailing unmatched parenthesis in conclusion-FOL
D35	(Knows(dan, dune) ∧ ScienceFiction(dune)) ∨ ProvedToBe(dune, false)) — Unmatched parenthesis in FOL formula — additional annotation error instance
WEB-6	2026 Agentified Assessment paper using Vampire theorem prover identified label errors and NL-FOL misalignments in original FOLIO; auto-formalization reached 86.70% on validation set after cleaning (https://arxiv.org/abs/2603.02788)

Information Gaps

- Total proportion of FOLIO examples affected by annotation errors observable only via systematic prover verification
- Whether the 2026 cleaned split is publicly accessible at time of deployment

Requires Expert Verification

- Logic-instructor spot-check of FOLIO labels against course rubric on a sample of HybLogic examples that match Hurley register

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Remediation

Gap 1: Observable annotation errors in the public HuggingFace release (free variables, name mismatches, unmatched parentheses) confirm a known systemic issue identified by the 2026 Vampire-prover cleaning study.

Recommendation: Prefer the 2026 cleaned FOLIO split [WEB-6] when available; if using the original release, run a syntactic check on all FOL formulas and manually verify a sample of False and Uncertain labels before reporting performance.

ID	Evidence
WEB-6	2026 Agentified Assessment paper using Vampire theorem prover identified label errors and NL-FOL misalignments in original FOLIO; auto-formalization reached 86.70% on validation set after cleaning (https://arxiv.org/abs/2603.02788)

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *FOLIO* | ***Context:*** *US Undergraduate Introductory Logic Course — Automated Validity Checker*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The deployment produces a single categorical label from a fixed three-item set (valid/invalid/indeterminate), and FOLIO's primary reasoning task evaluates a categorical label from True/False/Unknown with accuracy as the primary metric [Q80, Q66]. Output modality and form are well-matched. The NL-FOL translation metrics (SynV, ExcAcc [Q73–Q76]) are not deployment-relevant. Two minor wrinkles: (a) the public HuggingFace dataset does not expose the `test` split that published baselines [Q93, Q94] are computed on, limiting direct comparison; (b) accuracy alone (without per-class breakdown) is insufficient given documented model bias toward True predictions [Q155, Q156, Q157] — per-class accuracy should be reported. Literacy and accessibility are non-issues: the deployment population is US undergraduates in an English-language course.

ID	Evidence
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q73	“Two metrics are adopted to evaluate NL-FOL translation to capture different aspects of the generation results: 1). Syntactic validity (SynV).” (p.6)
Q74	“The Syntactic Validity score measures whether the FOL formulas are syntactically valid. The score will be 1 if all FOL formulas of an example can pass the syntactic check and 0 otherwise 2).” (p.6)
Q75	“Inference Engine execution accuracy (ExcAcc). The group of translated FOL for premises and conclusions in one story is fed into our inference engine to output the truth value for each conclusion.” (p.6)
Q76	“We define the accuracy of the output labels as the execution accuracy.” (p.6)
Q80	“We use accuracy for evaluating logical reasoning performance.” (p.6)
Q93	“We run experiments on five randomly sampled sets of examples from the training set and report the average accuracy.” (p.7)
Q94	“The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively.” (p.7)
Q155	“Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown.” (p.15)
Q156	“The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting.” (p.15)
Q157	“They also tend to make more predictions of True than the other labels.” (p.15)

Strengths

- Categorical label from fixed small set matches deployment exactly [Q66, Q80]
- Accuracy is a transparent, deployment-aligned metric
- No modality mismatch — text in, label out

ID	Evidence
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q80	“We use accuracy for evaluating logical reasoning performance.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (single categorical label) matches deployment need exactly [Q66, Q80].

OF-2: Text-to-speech not relevant; deployment is a text-based tool used by literate undergraduates.

OF-3: Literacy assumed (US undergraduate population); accessibility-of-output not a concern at this stage.

OF-4: Practical concerns: (a) test split unavailable on HuggingFace (DATASET limitation) — published baselines [Q93, Q94] cannot be directly replicated; (b) overall accuracy hides class-imbalance effects, so per-class accuracy should be reported [Q155, Q156].

ID	Evidence
Q66	“Given P and H, the goal is to determine the truth values of the conclusions: "True", "False" or "Unknown", based on FOL reasoning.” (p.6)
Q80	“We use accuracy for evaluating logical reasoning performance.” (p.6)
Q93	“We run experiments on five randomly sampled sets of examples from the training set and report the average accuracy.” (p.7)
Q94	“The majority baseline of our dataset is 38.5% since in our test set, there are 87, 78 and 61 examples with labels of true, false and unknown respectively.” (p.7)
Q155	“Confusion matrices in Figure 4 for the fine-tuning and 8-shot NL prompt results both show that LLMs are significantly better at making the correct predictions for conclusions with labels of True than the conclusions with labels of False or Unknown.” (p.15)
Q156	“The accuracy on examples with False or Unknown conclusions is 61.9% with fine-tuning and 54.0% with few-shot prompting.” (p.15)

Information Gaps

- Whether the public release of FOLIO test split has been updated since dataset analysis

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Remediation

Gap 1: Public HuggingFace release does not expose the test split that published baselines are computed on, limiting direct replication of GPT-4 64.16% / expert 95.98% comparisons.

Recommendation: Use the validation split (203 examples) for deployment-aligned benchmarking and document the split used; treat published test-set baselines as approximate references rather than exact targets.